

Mathematical analysis 2

Chapter 6 : Fourier series



2024/2025

Course outline

1 Generalities

2 Trigonometric series

3 Fourier series

Some definitions

Definition

We say that a function defined on a subset D of $\mathbb{R}$ is T -periodic if:

$$\forall x \in D: \quad x + T \in D \quad \text{and} \quad f(x + T) = f(x).$$

Lemma

- If f is T -periodic, then $f(x + nT) = f(x)$ for any integer n .
- The functions $\cos(mx)$ and $\sin(mx)$ have a period $T = \frac{2\pi}{m}$ ($m \neq 0$).

Lemma

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a T -periodic function where $T > 0$, integrable over the interval $[0, T]$. Then:

$$\forall \alpha \in \mathbb{R}, \quad \int_{\alpha}^{\alpha+T} f(t) dt = \int_0^T f(t) dt.$$

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Trigonometric series

Definition

- We call *trigonometric series* all series of functions $\sum f_n$ whose general term is of the form:

$$f_0(x) = \frac{a_0}{2} \quad \text{and} \quad f_n(x) = a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \quad \forall n \geq 1,$$

where $l \neq 0$, and $(a_n)_n$ and $(b_n)_n$ are two sequences called the *coefficients of the trigonometric series*.

- A *trigonometric series* is denoted as:

$$\frac{a_0}{2} + \sum_{n \geq 1} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

Trigonometric series

Example.

- The series $\sum_{n \geq 1} \frac{\cos(nx)}{n^2}$ is a trigonometric series with $l = \pi$,

$a_0 = 0$, and for all $n \in \mathbb{N}^*$, $a_n = \frac{1}{n^2}$ and $b_n = 0$.

- The series $2 + \sum_{n \geq 1} \frac{\cos(nx)}{n^2} + \frac{\sin(nx)}{n}$ is a trigonometric series

with $l = \pi$, $a_0 = 4$, and for all $n \in \mathbb{N}^*$, $a_n = \frac{1}{n^2}$ and $b_n = \frac{1}{n}$.

Convergence of trigonometric series

Theorem

If the numerical series $\sum a_n$ and $\sum b_n$ converge absolutely, then the trigonometric series:

$$\frac{a_0}{2} + \sum_{n \geq 1} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

converges normally (and then absolutely and uniformly) over $\mathbb{R}$.

Convergence of trigonometric series

Proposition

Let $(a_n)_n$ and $(b_n)_n$ be two sequences of positive real numbers, decreasing, and converging to 0. The trigonometric series:

$$\frac{a_0}{2} + \sum_{n \geq 1} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

- converges pointwise for all $x \in \mathbb{R}$ such that $x \neq 2kl$, $k \in \mathbb{Z}$.
- converges uniformly on every interval $[2kl + \alpha, 2(k+1)l - \alpha]$ where $k \in \mathbb{Z}$ and $\alpha \in]0, l[$.

Properties of the sum of a trigonometric series

Proposition (Continuity)

If the trigonometric series $\frac{a_0}{2} + \sum_{n \geq 1} (a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l})$ converges uniformly over an interval I , then its sum is a continuous function on I .

Proof.

- All functions f_n such that $f_n(x) = a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l}$ are continuous on $\mathbb{R}$;
- The series $\frac{a_0}{2} + \sum_{n \geq 1} (a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l})$ converges uniformly on I ; thus, the sum function is continuous on I .

Properties of the sum of a trigonometric series

Proposition (Periodicity)

If the trigonometric series $\frac{a_0}{2} + \sum_{n \geq 1} (a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l})$ converges, then its sum is a periodic function with period $2l$.

Suppose the series $\frac{a_0}{2} + \sum_{n \geq 1} (a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l})$ converges and its sum is equal to $S(x)$, i.e., $S(x) = \frac{a_0}{2} + \sum_{n \geq 1} (a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l})$. Let's demonstrate that $S(x + 2l) = S(x)$:

$$\begin{aligned} S(x + 2l) &= \frac{a_0}{2} + \sum_{n \geq 1} (a_n \cos(\frac{n\pi}{l}(x + 2l)) + b_n \sin(\frac{n\pi}{l}(x + 2l))) \\ &= \frac{a_0}{2} + \sum_{n \geq 1} (a_n \cos(\frac{n\pi x}{l} + 2n\pi) + b_n \sin(\frac{n\pi x}{l} + 2n\pi)) \\ &= \frac{a_0}{2} + \sum_{n \geq 1} (a_n \cos(\frac{n\pi x}{l}) + b_n \sin(\frac{n\pi x}{l})) = S(x) \end{aligned}$$

Links between the sum of a trigonometric series and the coefficients of the series

Proposition

If the trigonometric series: $a_0 + \sum_{n \geq 1} (a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l})$ converges uniformly to f on $[-l, l]$, then its coefficients are given by:

$$a_0 = \frac{1}{l} \int_{-l}^l f(x) dx$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx, \quad \forall n \geq 1$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx, \quad \forall n \geq 1$$

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Expansion of a function into a Fourier series

Given a function f from $\mathbb{R}$ to $\mathbb{R}$, $2l$ -periodic, we would like to find coefficients $(a_n)_n$ and $(b_n)_n$ such that f can be expanded into a trigonometric series, that is:

$$f(x) = \frac{a_0}{2} + \sum_{n \geq 1} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

The purpose of this section is to address the following two questions:

- What properties must the function f have for the coefficients $(a_n)_n$ and $(b_n)_n$ to exist?
- If $(a_n)_n$ and $(b_n)_n$ exist, what properties must the function f have for its trigonometric series to converge and for its sum to be equal to $f(x)$?

Fourier coefficients

Definition

Let f be a function that is $2l$ -periodic, integrable over any bounded closed interval in $\mathbb{R}$.

- The **Fourier coefficients** of f are defined as follows:

$$a_0 = \frac{1}{l} \int_{-l}^l f(x) dx$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx, \quad \forall n \geq 1$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx, \quad \forall n \geq 1$$

- The **Fourier series** associated with f is the trigonometric series given by:

$$Ff(x) = \frac{a_0}{2} + \sum_{n \geq 1} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

Remarks

Remark

- We can replace the boundaries of the integrals in the previous definition by $\int_0^{2l}$ or to any interval of length $2l$.
- It is not evident that $Ff(x)$ converges, and even if it converges, its sum is not necessarily $f(x)$.

Definitions

Definition

- A function f is said to be **even** if $f(x) = f(-x) \forall x \in D_f$.
- The graph of f is symmetric with respect to the y -axis.
- In this case, $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$ for all a .

Definition

- A function f is said to be **odd** if $f(x) = -f(-x) \forall x \in D_f$.
- The graph of f is symmetric with respect to the origin O .
- In this case, $\int_{-a}^a f(x) dx = 0$ for all a .

Fourier series of odd and even functions

Proposition

Let f be a function that is $2l$ -periodic and integrable over any closed and bounded interval in $\mathbb{R}$.

① If f is **even**, then for all $n \in \mathbb{N}^*$, $b_n = 0$ and

$$a_0 = \frac{2}{l} \int_0^l f(x) dx; \quad a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx, \quad \text{for } n \geq 1.$$

② If f is **odd**, then for all $n \in \mathbb{N}$, $a_n = 0$ and

$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx, \quad \text{for } n \geq 1.$$

Case of functions 2π periodic ($l = \pi$)

Proposition

Let f be a 2π -periodic function, integrable over any closed and bounded interval in $\mathbb{R}$.

- ① The Fourier coefficients of f are given by: $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx, \quad \forall n \geq 1.$$

- ② The Fourier series associated with f is:

$$Ff(x) = \frac{a_0}{2} + \sum_{n \geq 1} (a_n \cos(nx) + b_n \sin(nx))$$

- ③ If f is even, then for all $n \in \mathbb{N}^*$, $b_n = 0$ and $a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos(nx) dx, \quad \forall n \geq 1$$

- ④ If f is odd, then for all $n \in \mathbb{N}$, $a_n = 0$ and

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin(nx) dx, \quad \forall n \geq 1$$

Case of functions 2π periodic ($l = \pi$)

Example: Consider the function $f : [-\pi, \pi] \rightarrow \mathbb{R}$, a 2π -periodic and even function defined by $f(x) = x$ if $x \in [0, \pi]$.

- Plot the graph of f in the interval $[-2\pi, 2\pi]$.
- Calculate the Fourier coefficients of f and provide its Fourier series.

Solution

1) Graph of f in the interval $[-2\pi, 2\pi]$:

2) Calculation of the Fourier coefficients of f :

f is 2π -periodic over $\mathbb{R}$. f is continuous over $\mathbb{R}$, thus it is integrable over any bounded closed set of $\mathbb{R}$. Then Ff exists.

f is even, so $b_n = 0, \forall n \geq 1$. And

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} x dx = \frac{2}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} = \pi.$$

$$\forall n \geq 1, a_n = \frac{2}{\pi} \int_0^\pi f(x) \cos(nx) dx = \frac{2}{\pi} \int_0^\pi x \cos(nx) dx$$

By integrating by parts,

$$\begin{cases} u = x \\ v' = \cos(nx) \end{cases} \implies \begin{cases} u' = 1, \\ v = \frac{1}{n} \sin(nx) \end{cases}$$

$$a_n = \frac{2}{n\pi} [x \sin(nx)]_0^\pi - \frac{2}{n\pi} \int_0^\pi \sin(nx) dx = \frac{2}{n^2\pi} (\cos(n\pi) - 1) = \frac{2}{n^2\pi} ((-1)^n - 1).$$

We obtain:

$$Ff(x) = \frac{\pi}{2} + \sum_{n \geq 1} \frac{2((-1)^n - 1)}{n^2\pi} \cos(nx).$$

We notice that

$$a_n = \begin{cases} -\frac{4}{(2k+1)^2\pi}, & \text{if } n = 2k+1, k \in \mathbb{N} \\ 0, & \text{otherwise} \end{cases}$$

$$Ff(x) = \frac{\pi}{2} - \sum_{n \geq 0} \frac{4}{(2n+1)^2\pi} \cos((2n+1)x).$$

Piecewise continuity and piecewise differentiability

Definition

We say that the function f is **piecewise continuous** on $[a, b]$ if there exists a subdivision $a = x_0 < x_1 < \dots < x_i < \dots < x_n = b$ such that:

- For every i , f is continuous on each $]x_i, x_{i+1}[$,
- and $\lim_{x \rightarrow x_i^+} f(x)$ and $\lim_{x \rightarrow x_{i+1}^-} f(x)$ exist and are finite, $i := 0; \dots; n-1$.

Definition

We say that the function f is **C^1 piecewise** on $[a, b]$ if there exists a subdivision $a = x_0 < x_1 < x_2 < \dots < x_n = b$ such that:

- For every i , f is **C^1** on each $]x_i, x_{i+1}[$,
- and $\lim_{x \rightarrow x_i^+} f'(x)$ and $\lim_{x \rightarrow x_{i+1}^-} f'(x)$ exist and are finite, $i := 0; \dots; n-1$.

Dirichlet theorem

What properties must the function f possess for its Fourier series to converge and for its sum to be equal to f ?

Theorem

Let f be a function that is $2l$ -periodic, integrable over any bounded closed set in $\mathbb{R}$, and let $x_0 \in \mathbb{R}$ satisfy:

- ① $\lim_{x \rightarrow x_0^+} f(x) = f(x_0^+)$, and $\lim_{x \rightarrow x_0^-} f(x) = f(x_0^-)$, exist and are finite.
- ② $\lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0^+)}{x - x_0}$ and $\lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0^-)}{x - x_0}$ exist and are finite.

Then, $Ff(x_0)$ converges, and its sum is equal to $\frac{1}{2}(f(x_0^+) + f(x_0^-))$, i.e.,

$$Ff(x_0) = \frac{1}{2}(f(x_0^+) + f(x_0^-)).$$

Additionally, if f is continuous at x_0 (i.e., $f(x_0^+) = f(x_0^-) = f(x_0)$), then:

$$Ff(x_0) = f(x_0).$$

Dirichlet theorem

Corollary

If f is a $2l$ -periodic function, piecewise C^1 on $[-l, l]$, then the Fourier series Ff of f converges on $\mathbb{R}$, and its sum is given by:

$$\forall x \in \mathbb{R}, Ff(x) = \frac{1}{2} (f(x^+) + f(x^-))$$

Moreover, if f is continuous on $E \subseteq \mathbb{R}$, then: $\forall x \in E, Ff(x) = f(x)$.

Parseval formula

Theorem

Let f be a $2l$ -periodic function, integrable over any bounded closed set of $\mathbb{R}$, with Fourier coefficients $(a_n)_n$ and $(b_n)_n$. Then:

① $\sum_{n \geq 1} (a_n^2 + b_n^2)$ is a convergent numerical series.

② $\frac{1}{l} \int_{-l}^l f^2(x) dx = \frac{a_0^2}{2} + \sum_{n \geq 1} (a_n^2 + b_n^2)$ (Parseval's Formula).

Corollary

If f is a $2l$ -periodic function, integrable over any bounded closed set of $\mathbb{R}$, then its Fourier coefficients tend toward zero as $n \rightarrow +\infty$, which means that

$$\lim_{n \rightarrow +\infty} a_n = 0 \quad \text{and} \quad \lim_{n \rightarrow +\infty} b_n = 0$$

Example: Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$, a 2π -periodic and odd function defined by $f(x) = 1, \forall x \in [0, \pi]$.

- Plot the graph of f in the interval $[-2\pi, 2\pi]$.
- Calculate the Fourier coefficients of f and provide its Fourier series.
- Study the convergence of Ff .
- Derive the value of the series:

$$\sum_{n \geq 0} (-1)^n \frac{1}{2n+1}, \sum_{n \geq 0} \frac{1}{(2n+1)^2}, \sum_{n \geq 1} \frac{1}{n^2}, \sum_{n \geq 1} (-1)^{n+1} \frac{1}{n^2}.$$

Solution:

1. **The graph of f :**
2. **Calculating the Fourier coefficients of f :** f is 2π -periodic over $\mathbb{R}$. f is piecewise continuous over $\mathbb{R}$, hence it is integrable over any closed and bounded set of $\mathbb{R}$. Thus, Ff exists.

As f is odd, $a_n = 0, \forall n \geq 0$. And

$$\forall n \geq 1, b_n = \frac{2}{\pi} \int_0^\pi \sin(nx) dx = \frac{2}{\pi} \left[-\frac{1}{n} \cos(nx) \right]_0^\pi = -\frac{2}{n\pi} (\cos(n\pi) - 1)$$

Then

$$\forall n \geq 1, b_n = -\frac{2((-1)^n - 1)}{n\pi}.$$

Thus,

$$Ff(x) = \frac{a_0}{2} + \sum_{n \geq 1} (a_n \cos(nx) + b_n \sin(nx)) = \sum_{n \geq 1} \frac{2(1 - (-1)^n)}{n\pi} \sin(nx).$$

We can see that:

$$b_n = \begin{cases} \frac{4}{(2k+1)\pi}, & \text{if } n = 2k + 1, k \in \mathbb{N} \\ 0, & \text{otherwise} \end{cases}.$$

Hence,

$$Ff(x) = \sum_{n \geq 0} \frac{4}{(2n+1)\pi} \sin((2n+1)x).$$

3. **Study of the convergence of Ff :** Since f is odd and 2π -periodic, it suffices to apply the Dirichlet's Theorem on $[0, \pi]$ to f . f is of class C^1 piecewise over $[0, \pi]$. Indeed, f is C^1 on $]0, \pi[$ as it's constant. And

$$\lim_{x \rightarrow \pi^-} f'(x) = \lim_{x \rightarrow \pi^-} 0 = 0 \in \mathbb{R}, \quad \lim_{x \rightarrow 0^+} f'(x) = \lim_{x \rightarrow 0^+} 0 = 0 \in \mathbb{R}.$$

Thus, Ff converges for all $x \in \mathbb{R}$, and at points of continuity of f , its sum equals:

$$Ff(x) = \sum_{n \geq 0} \frac{4}{(2n+1)\pi} \sin((2n+1)x) = f(x), \quad \forall x \in \mathbb{R} - \{n\pi\}.$$

And at points of discontinuity of f , its sum equals:

$$\forall x \in \{n\pi\}, \quad Ff(x) = \frac{f(x+) + f(x-)}{2} = 0 \neq f(x).$$

Conclusion: Ff converges toward f on $\mathbb{R} - \{n\pi\}$.